School of Computer Engineering, KIIT Deemed to be University

Computer Networks Laboratory (CS39003)

Lab Assignment

Packet Analysis using Wireshark: Protocols, Ping and Traceroute

Name	Arkadeep Mondal	**Roll No.**	2405036
Section / Group	CS22	**Date of Lab**	24/9/2026
Full Marks	10	**Due Date**	

Objective

Capture your own traffic with Wireshark, identify the protocols involved, follow an HTTP exchange, then study the ICMP echo request/reply exchange behind `ping` and the TTL-based hop discovery behind `traceroute`, reading the details straight from the captured IP and ICMP headers.

Ground rules

- Every answer must come from **your own captures**. Quote frame numbers so your answers can be checked against your `.pcapng` files.
- Capture on the interface that carries your Internet traffic (Wi-Fi or Ethernet). Close browsers and downloads before the ping and traceroute captures.
- Run `ping` and `traceroute` in a Linux or macOS terminal (or a Linux VM in bridged mode). Windows: `ping -n 5 www.kiit.ac.in` and `tracert www.kiit.ac.in`.
- If `www.kiit.ac.in` does not reply on your network, substitute another reachable site (e.g. your institution's own domain) in both commands, and note which host you used.
- Identical captures or answer sheets will be treated as copying, and all involved will lose the marks.

Marks distribution

Part	Topic	Marks	Awarded
A	Getting started: protocols and HTTP GET / OK	2	
B	Ping and the ICMP echo request/reply exchange	4	
C	Traceroute and IP header fields	3	
D	Evidence and presentation	1	
	Total	**10**	

Part A: Getting Started with Wireshark (2 marks)

Task A. Start Wireshark capturing on your active interface. While it runs, open `http://neverssl.com` in a browser (it never redirects to HTTPS, so the HTTP exchange is easy to capture). Stop the capture and save it as `RollNo_intro.pcapng`.

A1. In the **unfiltered** packet list, name three different protocols from the Protocol column and say in one line what each was doing. **[0.5]**

DNS: Trying to get the IP address for the given domain name

NTP:Sync the clocks between the server and my client computer

ARP: ask the mac address of a local ip , as in which device has been assigned a certain ip

A2. Apply the filter `http`. Find the GET request and the 200 OK reply. Give

(i) both frame numbers, and each packet's value in Wireshark's **Time column** (seconds since capture start, as shown, to at least 3 decimal places); then state the reply time minus the request time as the elapsed time;

GET- frame no 110 , time - 12.373

Reply OK- frame no 123 , time - 12.695

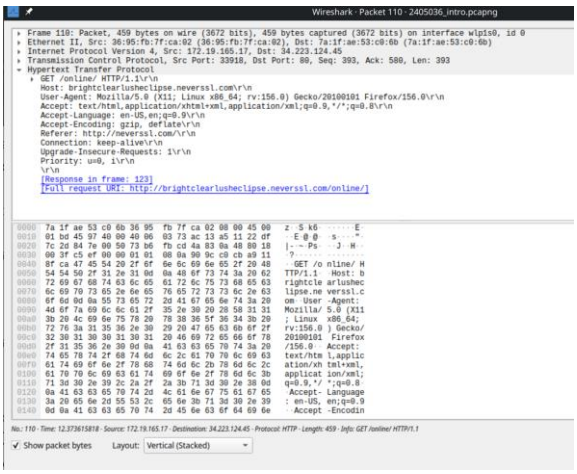
Elapsed Time: 0.322 sec

(ii) the IP address of `neverssl.com` and of your computer, read from the IP layer of the GET frame;

Local Ip : 172.19.165.17

Ip of neverssl : 32.223.124.45

(iii) a screenshot of the GET packet with Frame, Ethernet, IP and TCP **collapsed** and HTTP **expanded**. **[1.5]**



Part B: Ping and the ICMP Echo Exchange (4 marks)

Task B. Start a new capture, run the command below, then stop and save as `RollNo_ping.pcapng`.

```
ping -c 5 www.kiit.ac.in
```

B1. How many ICMP packets were exchanged in total? Give the filter you used, and split the count into requests sent and replies received. What ICMP **type and code** does a request use, and what type and code does a reply use? **[1]**

Ans:

ICMP Packets sent = 5 Packets

ICMP Packets Recieved = 5Packets

Total 10 Packets

Filter Used : ICMPv6

ICMP Request:

Type: Echo ping request 128

Code : 0

ICMP REPLY:

Type: Echo ping reply 129

Code : 0

B2. Complete **Table B-1** for all five pings. **[2]**

B3. In 4 to 6 lines, explain:

(a) how Wireshark (or `ping`) matches each reply to its request, given that several requests are in flight using the same protocol;

The ping replies are properly matched using the id and seq number fields in the ICMP header

Here the id is 0xdc3d and the sequence number goes from 1- 5

(b) why the Identification field differs between consecutive requests;

The id remains same but the seq field increases by 1 on each consecutive ping request ,this is done to properly match the reply packet with its request packet

(c) what the round-trip time tells you, and why it might vary between the five pings. **[1]**

The RTT time tells us about the latency between sending and receiving a packet. RTT varies across different pings as there are variations in network congestion , delays etc in between the intermediate routers, switches and the target server.

Table B-1: Ping requests and replies

Seq	Request frame no.	Request Time column value	Request TTL	Reply frame no.	Reply Time column value	Round-trip time (reply time – request time)
1	5	0.064903901	64	6	0.108230856	0.043326955
2	9	1.066308462	64	10	1.122897027	0.056588565
3	11	2.068234305	64	12	2.127985741	0.059751436
4	13	3.070510425	64	14	3.116122800	0.045612375
5	15	4.072765085	64	16	4.129446682	0.056681597

"Time column value" is the figure shown in Wireshark's Time column for that frame (seconds since capture start, as displayed, to at least 3 decimal places) - do not convert it to a clock time. Write "no reply" if a request got none.

Part C: Traceroute and IP Header Fields (3 marks)

Task C. Start a new capture, run the command below, then stop and save as `RollNo_traceroute.pcapng`. Keep the terminal output.

```
traceroute -q 5 www.kiit.ac.in
```

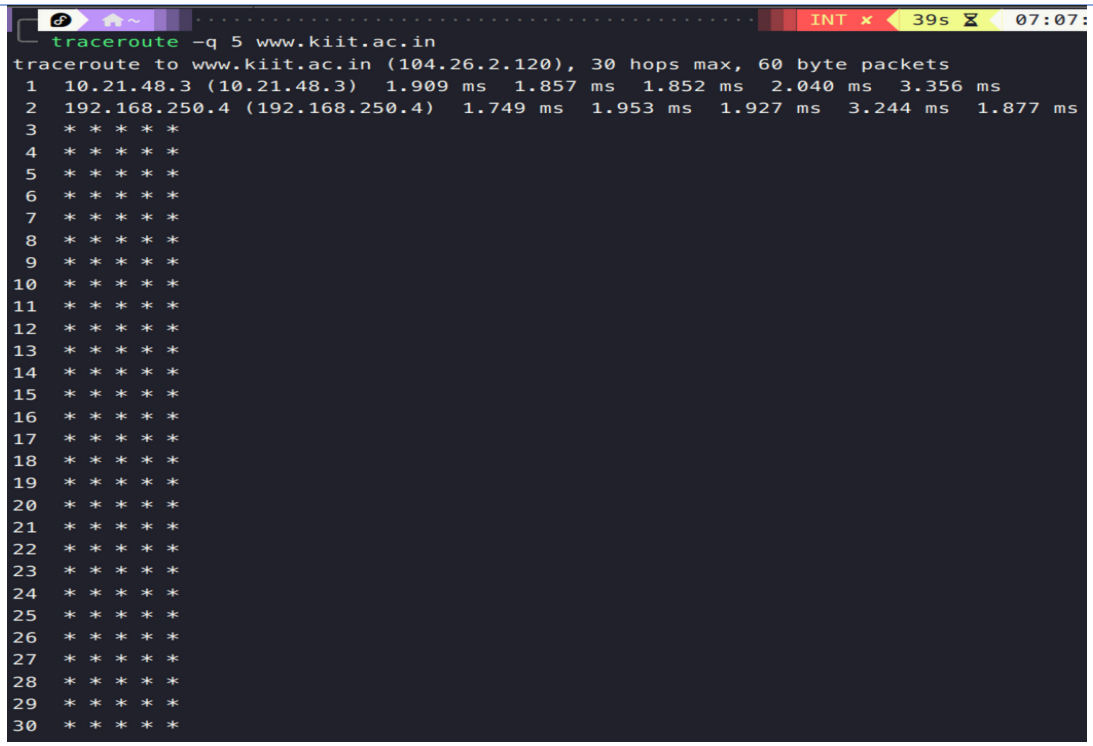
C1. How many hops did the traceroute show, and did it reach the destination or end in * lines? How many packets were sent by your client in total, and how many replies came back? Fill **Table C-1** (one row per router or server, plus the client row). Attach a screenshot of the terminal output. **[1.5]**

Number of hops: 30 hops

No it did not reach the final destination and ended in *** , some firewall blocked or dropped the UDP probe packets

Packets Sent: 150(30 X 5 probes due to –q 5)

Packets received : 10 (5 from hop 1 and 5 from hop 2)



C2. Across the probes your client sent, classify each IP header field below as **constant** or **changing**, and explain in one line why the protocol requires (or merely happens to produce) that behaviour. Fill **Table C-2**. **[1.5]**

Table C-1: Packet count per node

Hop	Router / server IP	Probes sent to this node	Replies received from this node	ICMP type of reply (if any)
1	10.21.48.3	5	5	Type: Time-to-live exceeded (11)
2	192.168.250.4	5	5	Type: Time-to-live exceeded (11)
3	No reply (* * * * *)	5	0	None(Timeout)
4	No reply (* * * * *)	5	0	None(Timeout)
5-30	No reply (* * * * *)	130	0	None(Timeout)
Client	**Your host (totals)**	150	10	

Table C-2: IP header fields across the traceroute probes

Field	Constant or changes?	Reason (one line: why the protocol requires or merely produces this)
Version	Constant	All outgoing probes exclusively use IPv4 (version 4) throughout the trace.
Total length	Constant	Each probe packet maintains a fixed total payload and header length (60 bytes).
Identification	Changes	The OS IP stack increments packet Identification to uniquely label each outgoing packet.
Time to live	Changes	Traceroute intentionally increments TTL by 1 after every set of 5 probes to discover successive network hops.

Field	Constant or changes?	Reason (one line: why the protocol requires or merely produces this)
Protocol	Constant	Standard Linux traceroute uses UDP (protocol number 17) for all outgoing probe packets.
Header checksum	Changes	Changing header fields like TTL and Identification means we have to recalculate the IP checksum
Source and destination address	Constant	All probes originate from local client IP (10.21.60.132) and target the same destination IP (104.26.2.120)

Part D: Evidence and Presentation (1 mark)

Submit the following as instructed by your lab instructor.

✓	Item	Marks
	Three capture files `RollNo_intro.pcapng`, `RollNo_ping.pcapng`, `RollNo_traceroute.pcapng`, all opening correctly in Wireshark	0.5
	One PDF report `RollNo_CN_Lab.pdf` with answers, tables and screenshots. Screenshots show frame numbers and the Time column and are labelled with the question number	0.5